

Game Analytics Tool

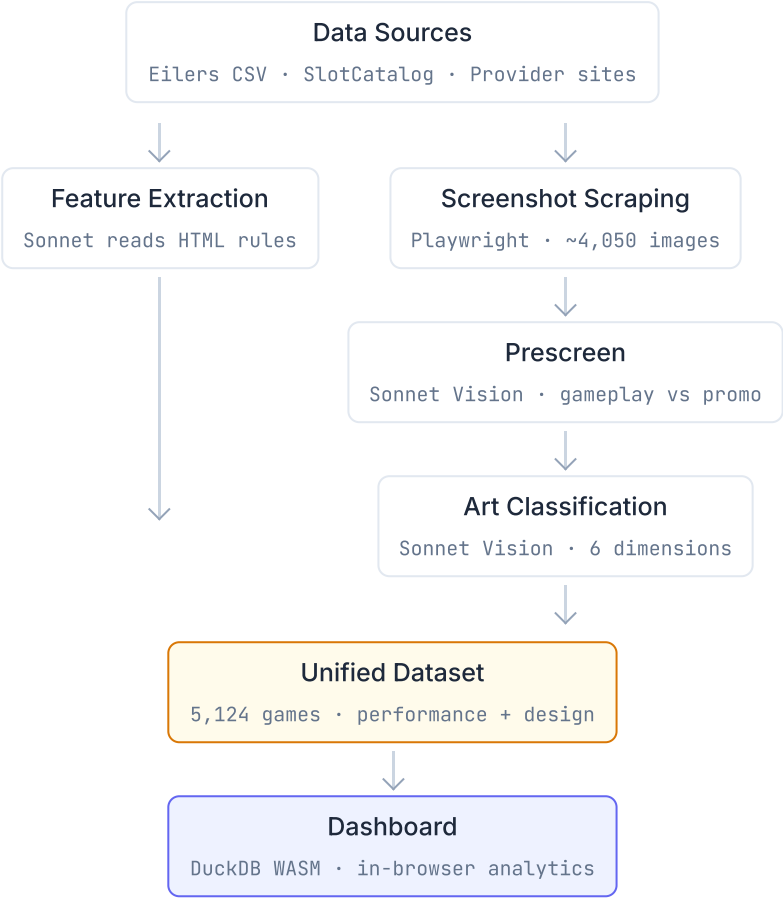
Using Claude Vision to connect game performance data with structured visual and mechanics data - across 5,124 games.

Problem: 5,124 games have performance data but no structured picture of how they look or play. Design decisions rely on gut feeling.

Approach: Claude Vision classifies screenshots across 6 art dimensions. Mechanics extracted from game rules pages. Everything linked to performance data.

Impact: Game designers make data-driven art and mechanics decisions based on what actually performs - replacing gut feeling and cutting research time significantly.

Pipeline Overview



The Game Analytics Tool

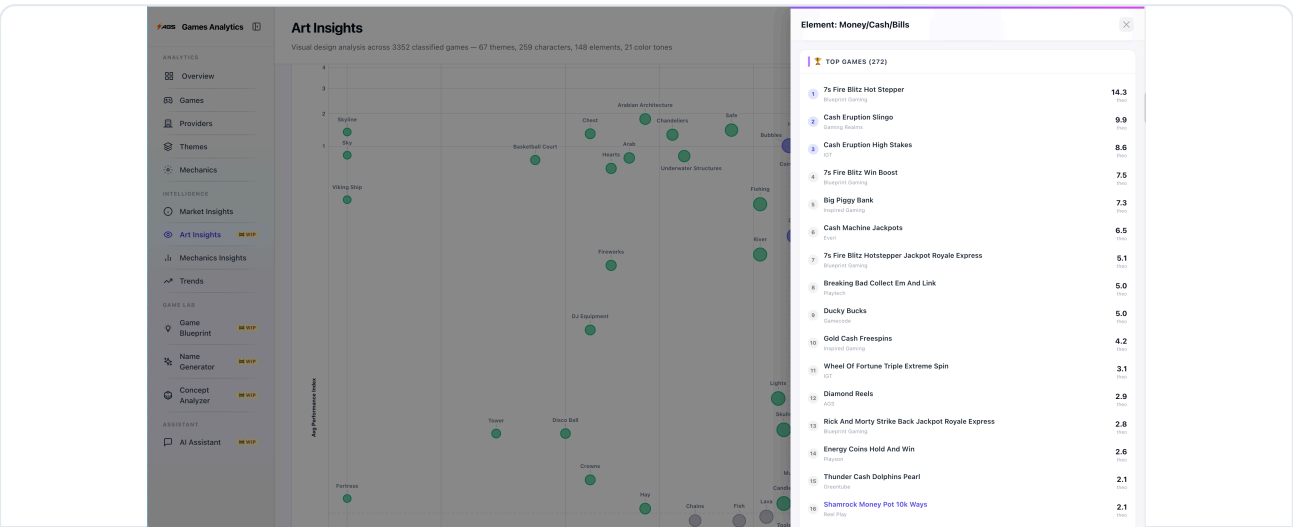
How we use it

Every art and mechanics decision is backed by performance data. A designer picks any dimension - theme, element, color, mechanic - and instantly sees which choices correlate with top-performing games across the full dataset.

The Elements landscape shows all game elements (Dragons, Gold Coins, Fire, etc.) as bubbles - size is how many games use it, position is performance vs. popularity:



Clicking it opens the full list - every game with that element, ranked by theo index. Each row shows the game name, provider, and performance score. From here you can drill into any game for its full classification, mechanics, and art data:



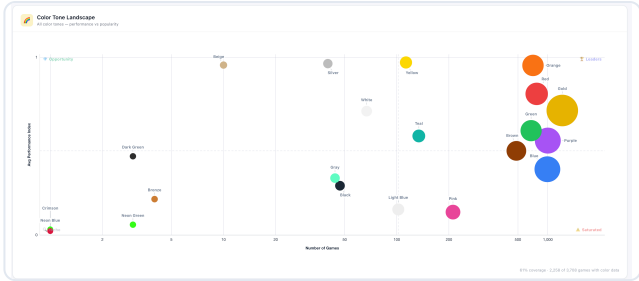
272 games ranked by theo index. 7s Fire Blitz Hot Stepper leads at 14.3, with provider and mechanic detail per game.

Every dimension has its own landscape chart - themes, characters, elements, colors, mechanics, narratives - performance vs. popularity at a glance:

Themes



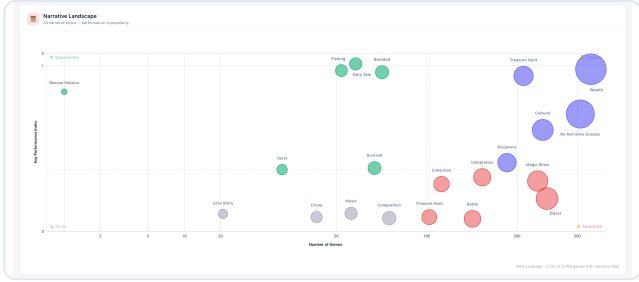
Colors



Characters



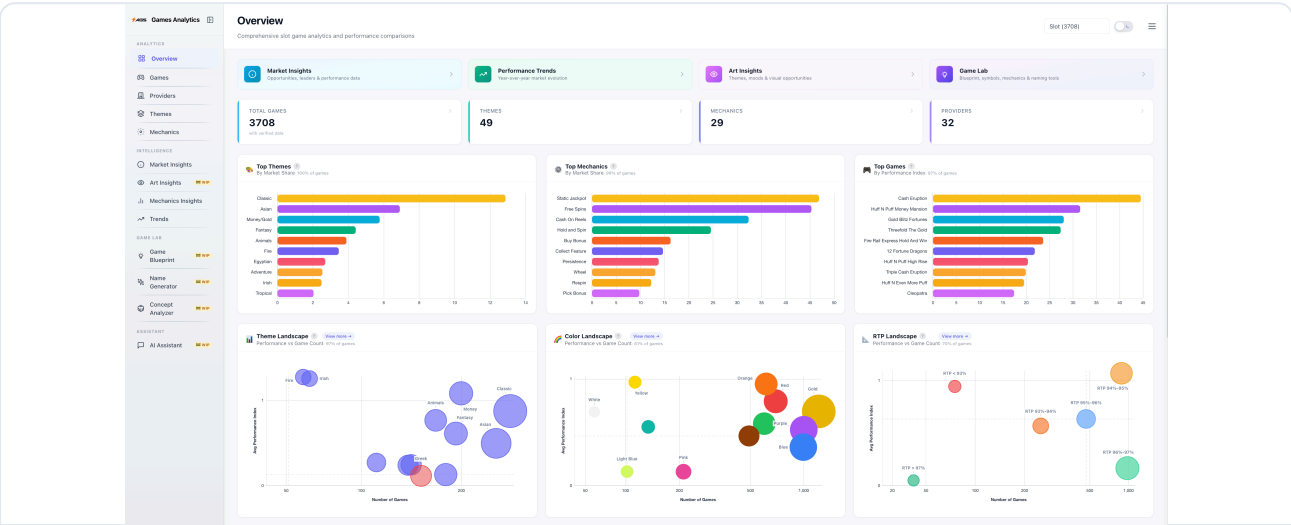
Narratives



All six dimensions have their own landscape chart - performance vs. popularity at a glance.

Overview and Landscapes

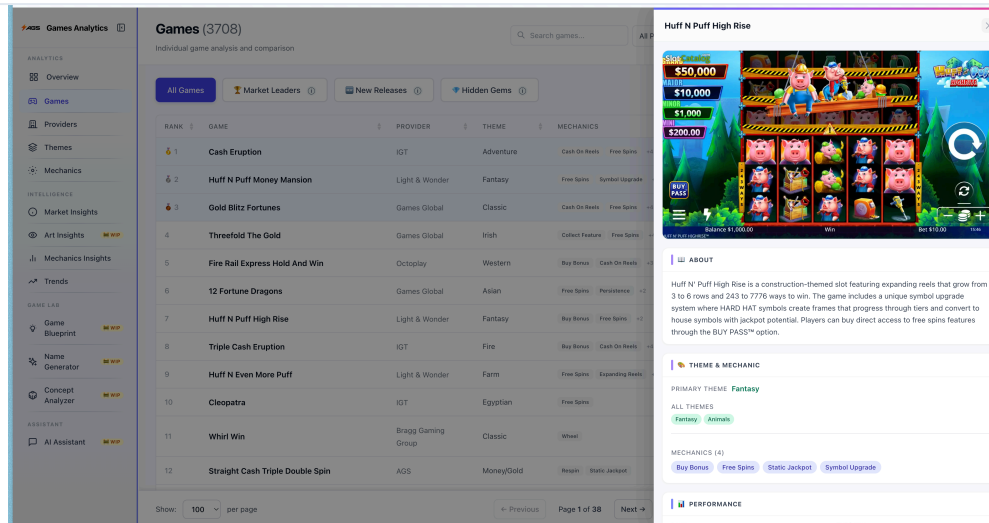
Landing page with KPIs, top themes/mechanics/games by Landscape market share, and landscape bubble charts for quick orientation.



3,708 games with verified data. Top themes, mechanics, and games ranked by performance.

Game Detail

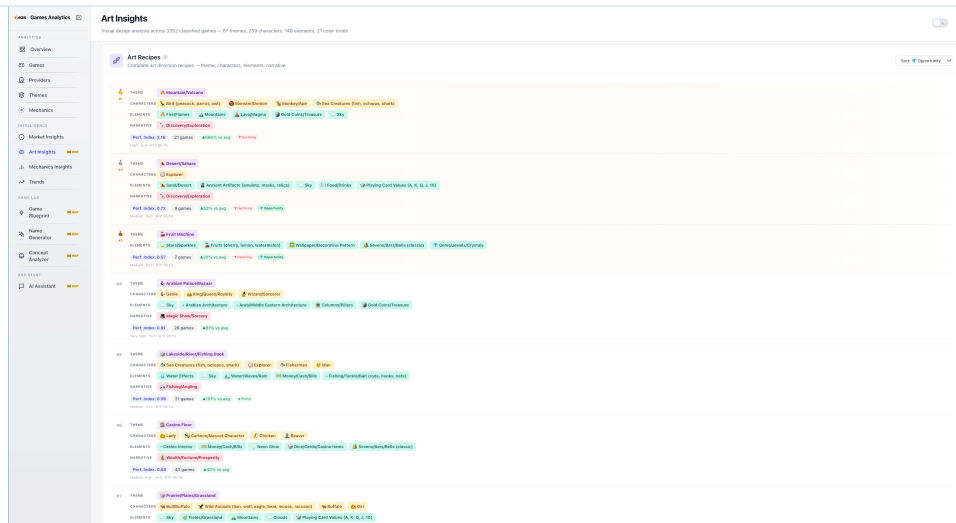
Click any game to see its full profile - classified art dimensions, mechanics, performance index vs. theme average, and how it compares to competitors. Every data point is queryable.



Game detail panel - scraped screenshot alongside classified theme, mechanics, and performance data

Art Recipes

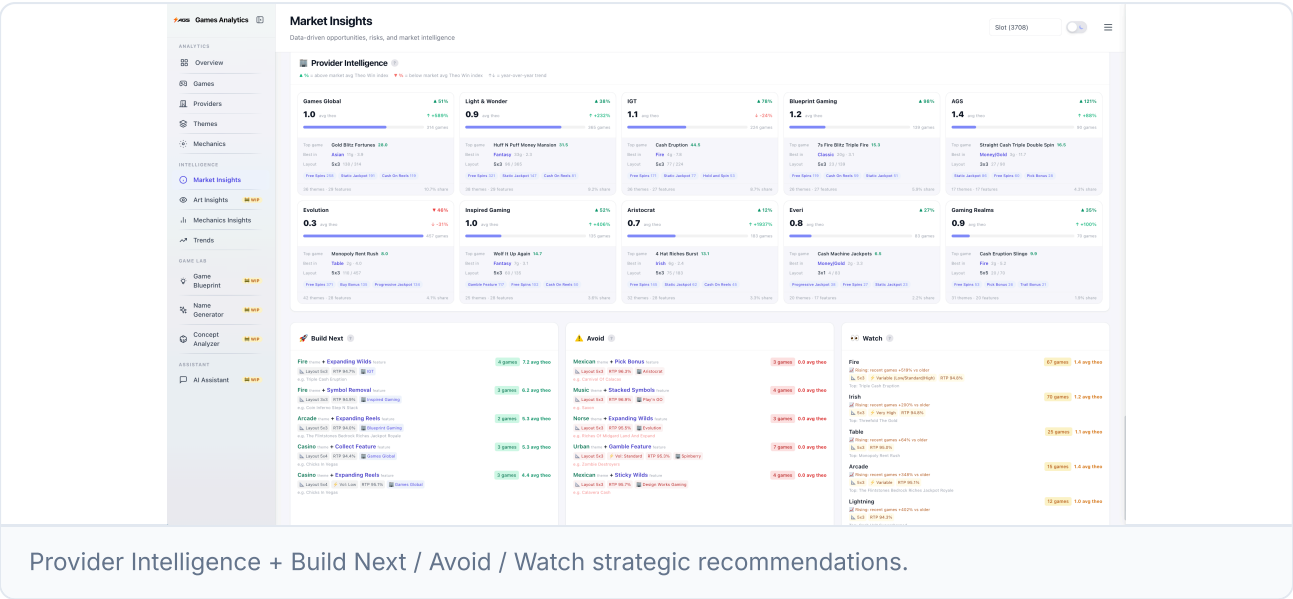
Full art direction breakdowns - theme, characters, elements, narrative - for top-performing games.
Low game count with high performance means untapped potential.



Art recipes: full visual breakdown per theme - characters, elements, narrative, performance index, and how it compares to average.

Market Intelligence

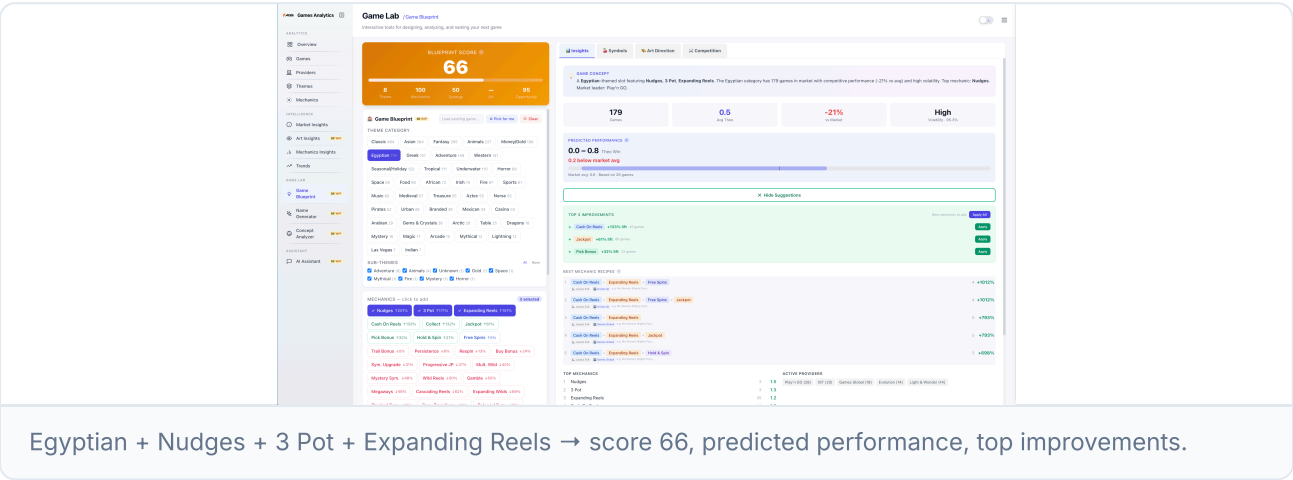
Provider cards showing each studio's top games, mechanics, and trend. Below: "Build Next" (high-opportunity combos), "Avoid" (saturated), and "Watch" (rising trends) - all derived from actual performance data across the full dataset.



Provider Intelligence + Build Next / Avoid / Watch strategic recommendations.

Game Blueprint

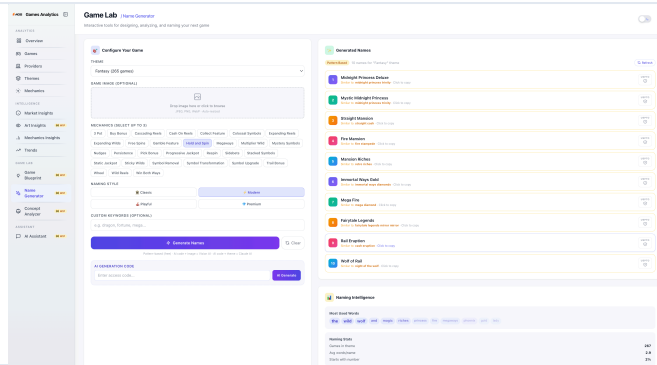
Pick a theme and mechanics to design a new concept. The tool scores it against real performance data from 5,124 games, predicts performance, and suggests data-backed improvements. The tabs below show art direction, recommended symbols, and competitive analysis - all for your selected configuration.



Egyptian + Nudges + 3 Pot + Expanding Reels → score 66, predicted performance, top improvements.

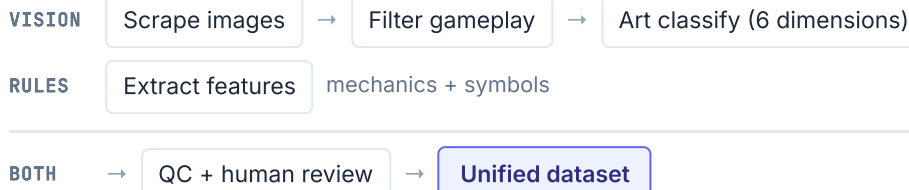
Game Name Generator

Analyzes naming patterns across 5,124 games - word count, keywords, structure - mapped to actual performance data. Suggests names based on what statistically correlates with higher-performing games in a given theme or mechanic.



Name generator: theme + mechanic input, pattern analysis, scored suggestions based on market data.

How We Built It



DATA COLLECTED VIA PLAYWRIGHT AUTOMATION

Gameplay screenshots	SlotCatalog, BigWinBoard, provider sites	~4,050 images
Game rules pages	BetMGM, Borgata, Play'n GO, Evolution, High 5	~8,800 pages
Performance data	Eilers CSV exports	5,124 games

Standard rate-limiting; most sites don't actively block. Same framework extends to lobby/placement scraping for competitive intelligence.

Screenshot Scraping

The art classification pipeline needs in-game screenshots as input - actual gameplay with the reel grid visible. These don't exist in any structured dataset, so we scrape them. ~4,050 gameplay screenshots collected from multiple sources via Playwright automation:

1 Scrape

Playwright visits SlotCatalog, BigWinBoard, provider sites. Downloads all candidate images.

2 Filter

Haiku Vision picks the gameplay image from banners, logos, and thumbnails on the page.

3 Prescreen

Sonnet Vision classifies each saved image: gameplay, promotional banner, or rules page. Only gameplay passes to art classification.

Art Classification with Claude Vision

Dual-Image Classification

We send Claude Vision two versions of each screenshot in one API call - full gameplay and a masked version with the reel grid blacked out:

IMAGE 1 - FULL GAMEPLAY

Feeds: **Theme · Colors · Characters · Mood**

The model sees the full game screen.

IMAGE 2 - REEL GRID BLACKED OUT

Feeds: **Background Elements only**

Reel area (18%-82% of frame) removed.

Only actual scenery remains.

Why: Without masking, the model confuses reel symbols with background scenery. A Viking ship on the reels is a *symbol*, not a scene element - but it looks identical to a real ship in the background.

IS / IS NOT Classification Rules

Each dimension has explicit boundary rules. The model classifies into a constrained vocabulary - 53 themes, 26 colors, 134 character terms, 125+ elements - queryable across all games. Six dimensions total (theme, colors, characters, elements, mood, narrative); the first four undergo human review.

EXAMPLE · THEME CLASSIFICATION

IS "CASINO FLOOR"

Games visually set ON a casino floor - you see card tables, roulette wheels, casino interior, chips, dealers. The setting is inside a casino.

IS NOT

Games about money/cash/gold that have no actual casino interior. Games with multiplier symbols. "Cash" in the name does not mean Casino Floor.

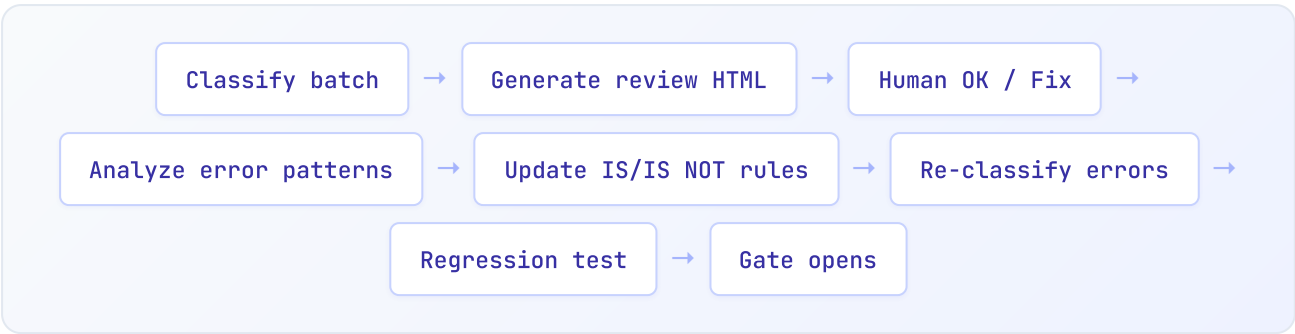
Roulette table visible, dealer character, casino chips → "Casino Floor"

Gold coins, treasure chest, "Cash Blast" title → NOT Casino Floor (likely Luxury or Classic Slots)

Every dimension has multiple rule cards like this - themes have one per category, characters have 6 distinct rule blocks. All grew iteratively from real classification errors.

Quality Control

Every batch goes through human review - 50+ rounds across the project. Recurring errors become new IS/IS NOT rules. A code-enforced gate blocks production until regression passes.



Multi-Dimension Review

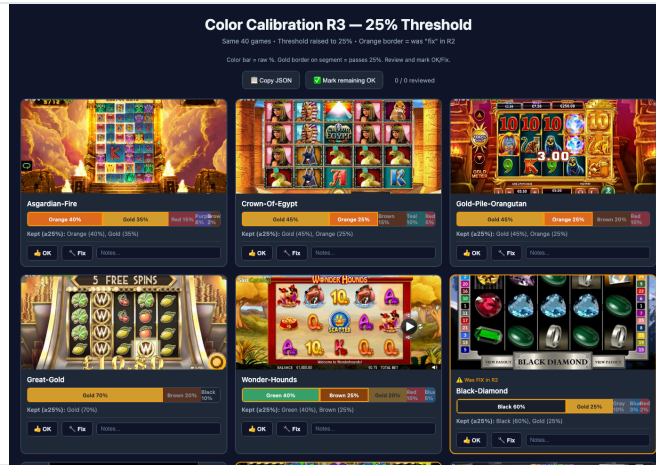
Each dimension (theme, colors, characters, elements) marked independently. Fix notes feed directly into prompt improvements.

The screenshot shows a 'Spot-Check' interface for 'Art Classification Batch 24'. It displays a slot machine game titled 'Astrocat' with a grid of symbols. Below the game, there are four dimensions for review: Theme (Outer Space), Colors (Purple, Gold, Blue), Characters (Cat), and Elements (Stars, Planets, Clouds). Each dimension has a 'Check' button and a 'Fix' button. The interface also shows 'Games reviewed: 0 / 20'.

Per-game review: screenshot + 4 dimensions, each independently marked OK/Fix with notes.

Color Calibration

The reviewer compares the model's percentage estimates against what they see on screen. Three rounds on 40 stratified games drove color accuracy from 77% to 82.5%.



Round 3 at 25% threshold. Color bars = raw model output. Gold borders = passed threshold. Orange cards = previously incorrect.

Feature Extraction + Symbols

In parallel with the screenshot pipeline above, we extracted mechanics and symbols from game rules pages.

1 Scrape rules pages

Indexes BetMGM / Borgata help pages, downloads full HTML (~8,800 pages). Strict title-matching links each to a game. Separate scrapers for Play'n GO, Evolution, High 5 Games. ~4,000 pages matched.

2 Game mechanics

Claude reads each page (text-only, no vision) and classifies into a 30-feature vocabulary (Hold and Spin, Free Spins, Megaways, etc.) with IS/IS NOT rules per mechanic.

3 Symbol names

Full payable extraction from the same pages - wilds, scatters, bonus triggers, themed and card symbols. Each typed with function descriptions.

Same IS/IS NOT approach as art classification - each mechanic has boundary rules so Claude knows exactly what counts. 30 mechanics, each with its own rule card.

EXAMPLE · HOLD AND SPIN

IS

Special symbols LOCK on the grid, remaining positions RESPIN with a counter that resets on new locks.

IS NOT

Regular respins. Free spins with sticky wilds. Vague "hold-style" without lock-grid- respin-counter.

3,674 games extracted at **97% accuracy** against 228 ground-truth games. Symbol names are also used as hints in the art pipeline - preventing Claude Vision from confusing reel symbols with actual characters.

Model Cascade

Different Claude models for different stages, based on task complexity vs. cost:

STAGE	MODEL	COST	WHAT IT DOES
Download filter	Haiku (vision)	\$0.0001/img	During Playwright scraping, pages have multiple images. Haiku picks the gameplay screenshot vs. promo banners or logos.
Screenshot prescreen	Sonnet (vision)	\$0.003/img	Classifies each saved screenshot: is this actual gameplay, a promotional banner, or a rules page? Only gameplay passes to art classification.
Art classification	Sonnet (vision, cached)	\$0.008/game	Full 6-dimension visual classification from the gameplay screenshot (dual-image, IS/IS NOT cards, constrained vocabulary).
Feature extraction	Sonnet (text-only)	\$0.005/game	Reads HTML game rules pages and extracts mechanics (Hold and Spin, Megaways, Free Spins, etc.) to a 30-feature vocabulary.

Prompt caching saves ~90% on system prompts. Batch API gives 50% off bulk runs. Checkpointed after every game.

Current Status

DIMENSION	ACCURACY	STATUS
Feature extraction	97%	Production
Screenshot prescreen	97.4%	Production
Art theme	92%	Production
Art colors	82.5%	Calibrated
Art characters	90%	Production
Art elements	~65%	Known vision limitation

Elements remain hardest - spatial reasoning for background vs. reel content is a known vision model limitation.

What's Next

NOW - IN PROGRESS

Full coverage reclassification

Latest prompt across all 5,124 games for unified art data.

Feature combination analysis

Which mechanic + art combinations drive performance together.

NEXT - PLANNED

Game naming

Naming pattern analysis - word count, keywords, structure mapped to performance.

Poster art

Layout, typography, color analysis for promotional materials.

Game types

Structural classification (concatenation, stepper, cluster).

5,124 games

97% feature accuracy

92% art theme

\$0.008/game

1,698 tests

50+ review rounds

Claude Vision (Anthropic) · DuckDB WASM · Playwright · Custom pipeline orchestration